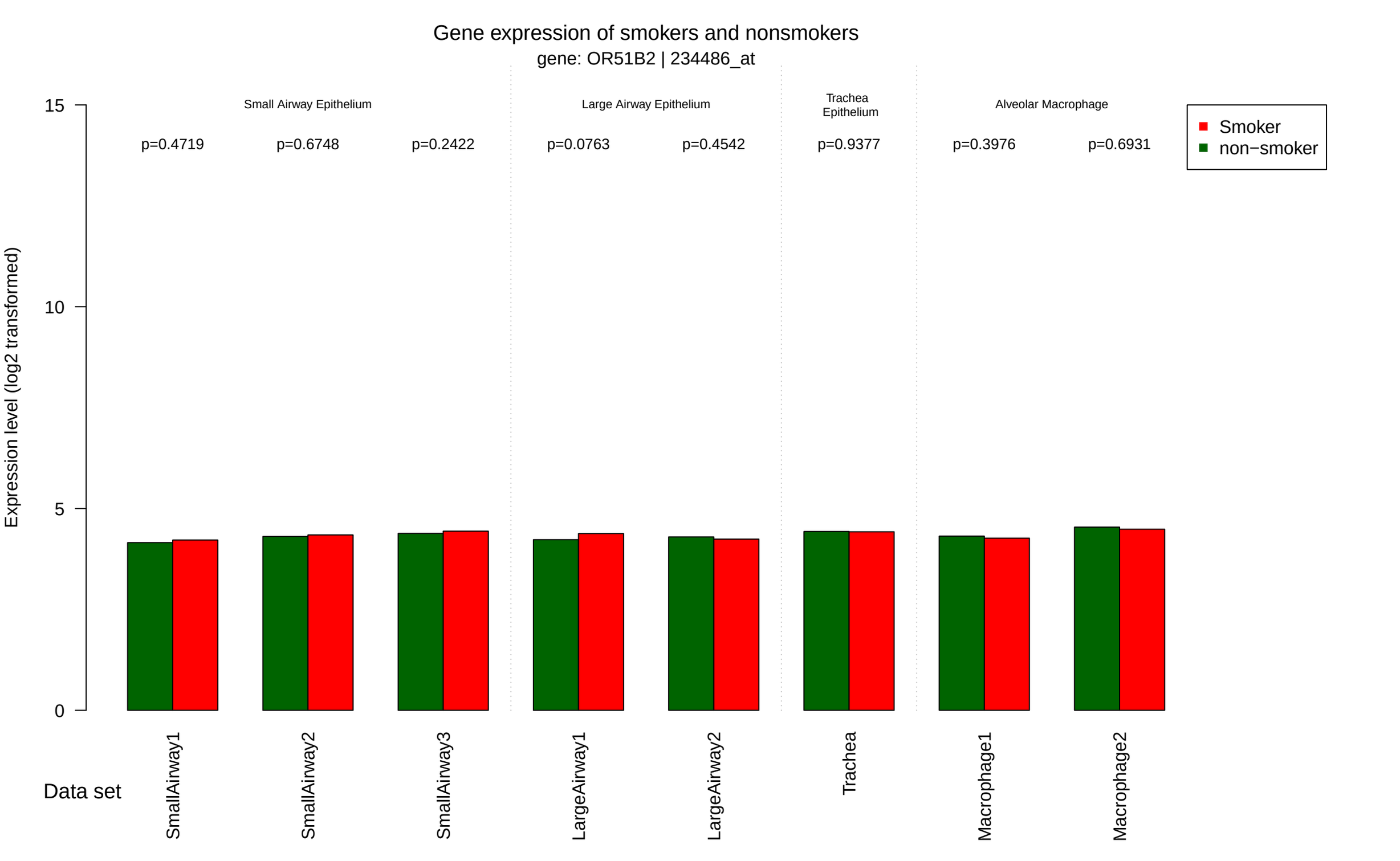
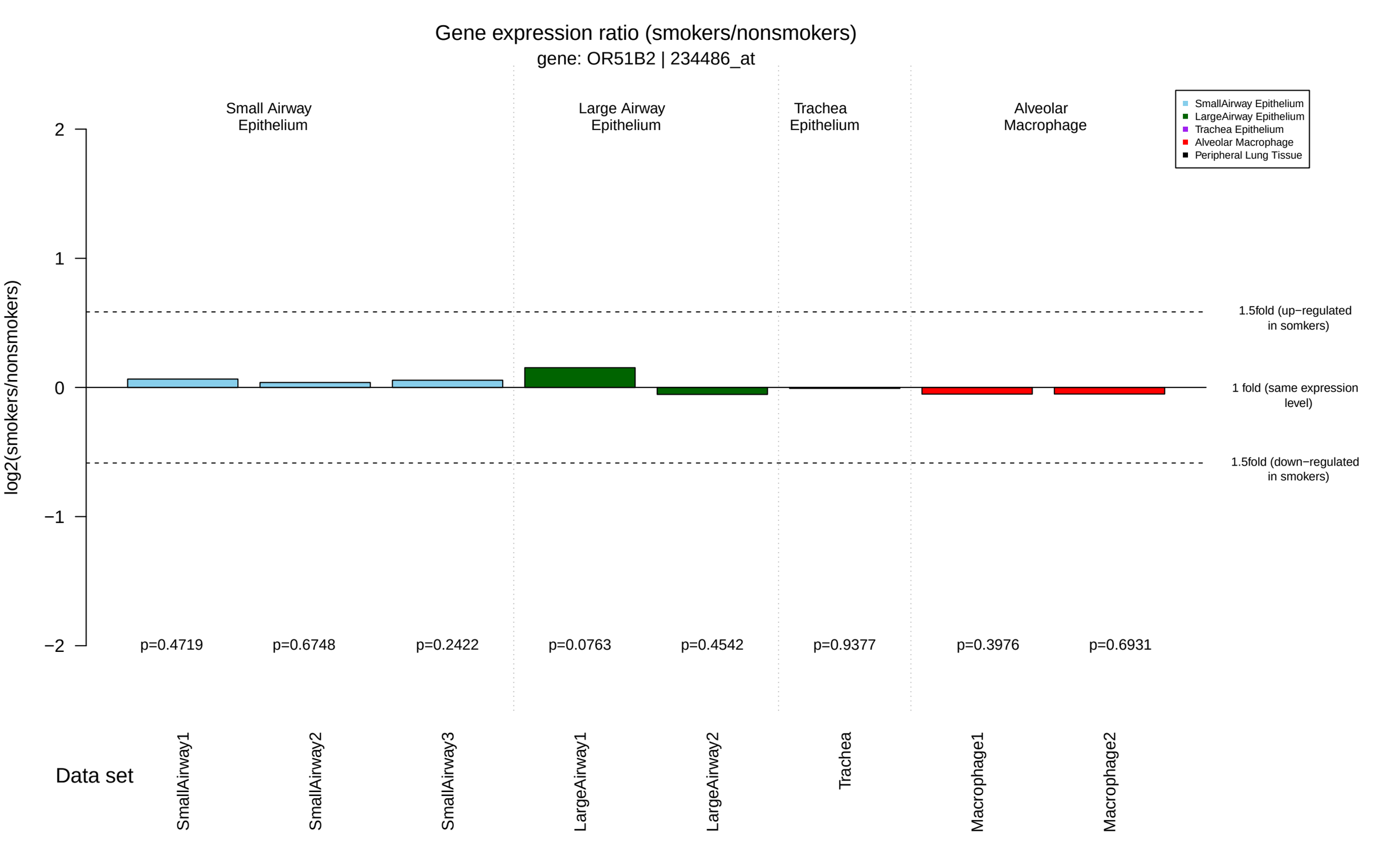
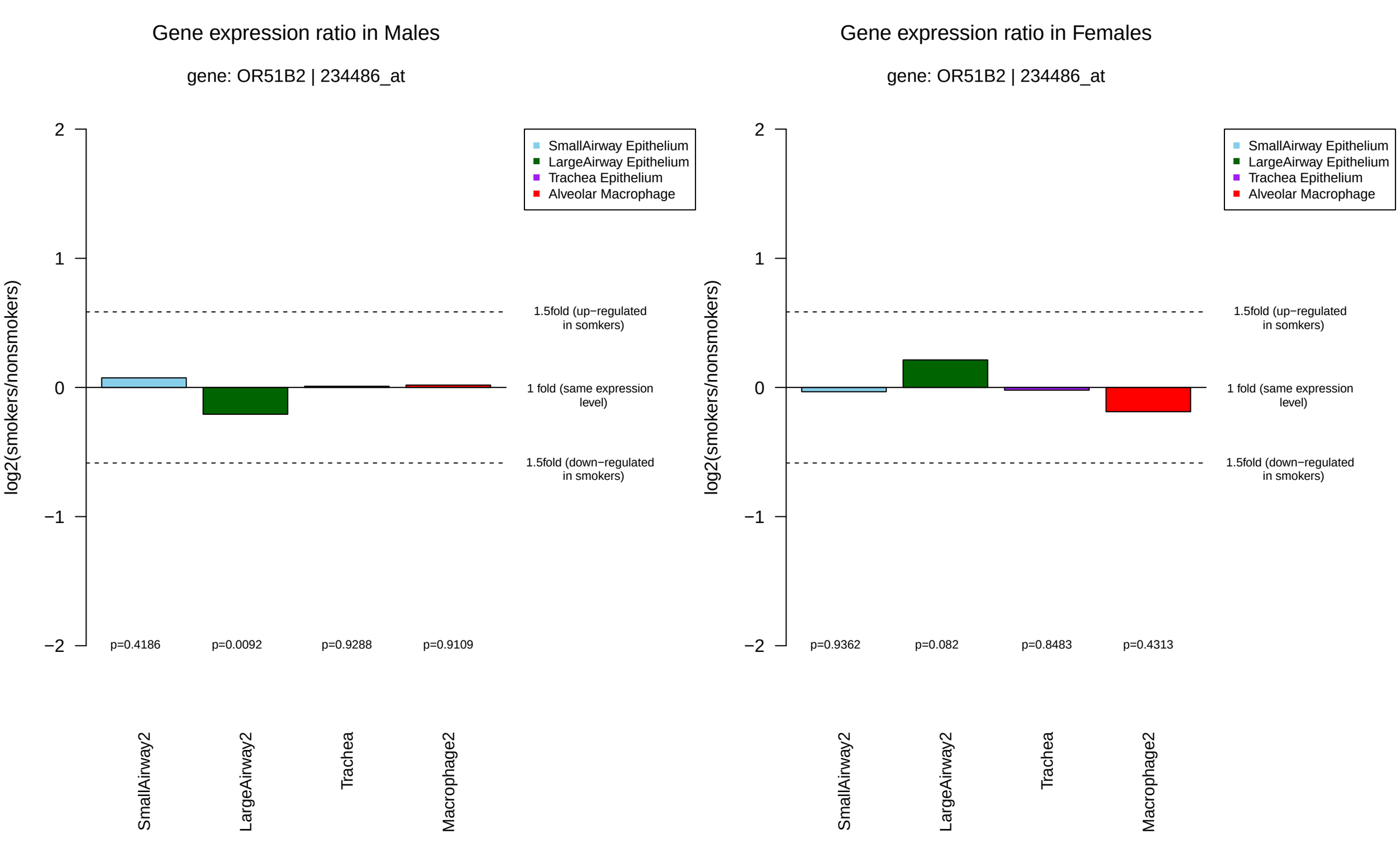
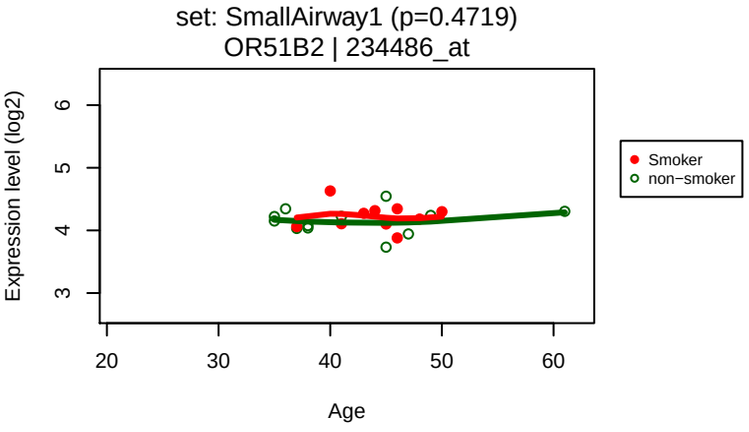
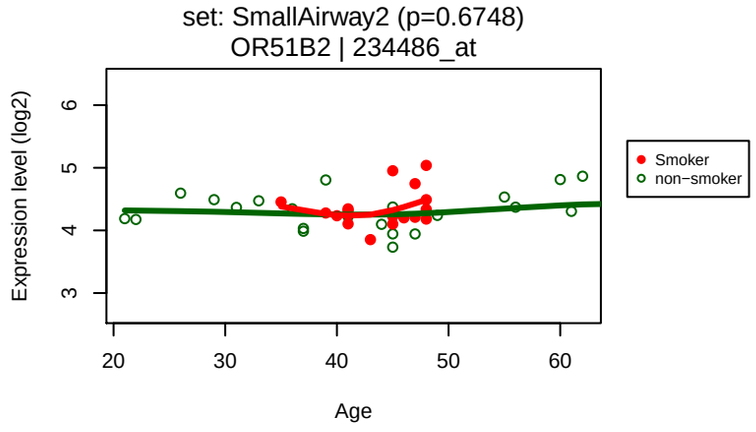
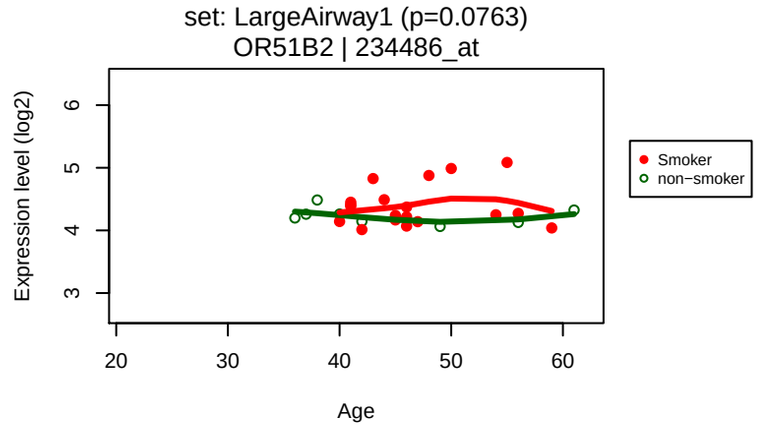

Gene expression of smokers and nonsmokers
gene: OR51B2 | 234486_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4719
p=0.6748
p=0.2422
p=0.0763
p=0.4542
p=0.9377
p=0.3976
p=0.6931
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

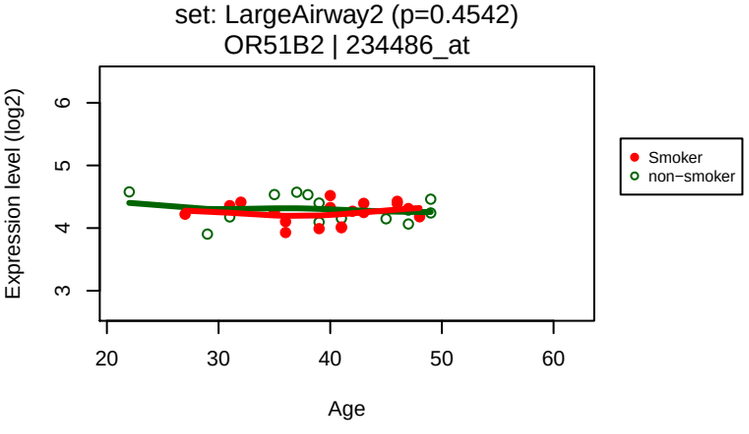

Gene expression ratio (smokers/nonsmokers)
gene: OR51B2 | 234486_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4719
p=0.6748
p=0.2422
p=0.0763
p=0.4542
p=0.9377
p=0.3976
p=0.6931
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

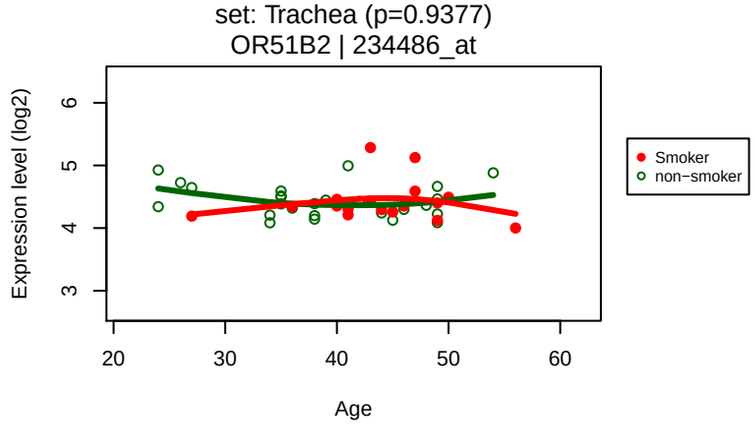

Gene expression ratio in Males
gene: OR51B2 | 234486_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4186
p=0.0092
p=0.9288
p=0.9109
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OR51B2 | 234486_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9362
p=0.082
p=0.8483
p=0.4313
SmallAirway2
LargeAirway2
Trachea
Macrophage2

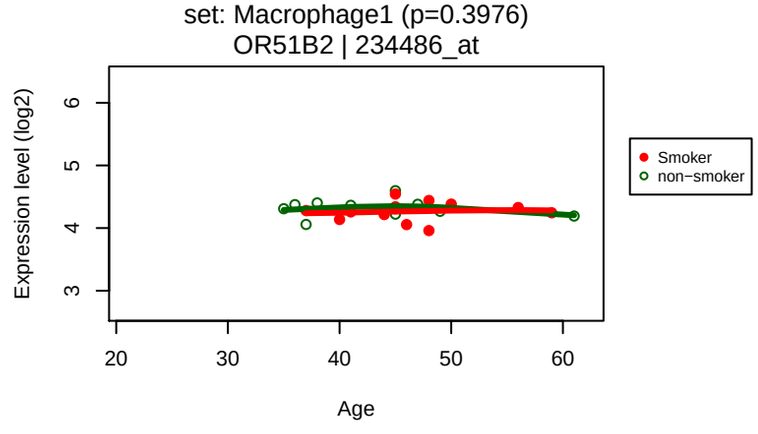

set: SmallAirway1 (p=0.4719)
OR51B2 | 234486_at
Expression level (log2)
Age
Smoker
non−smoker

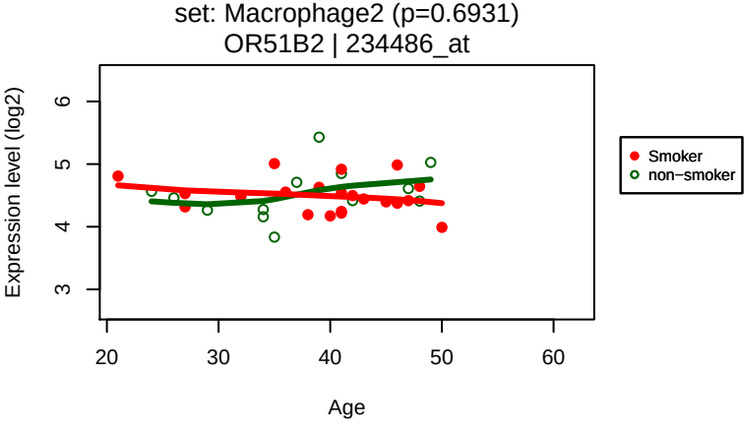

set: SmallAirway2 (p=0.6748)
OR51B2 | 234486_at
Expression level (log2)
Age
Smoker
non−smoker

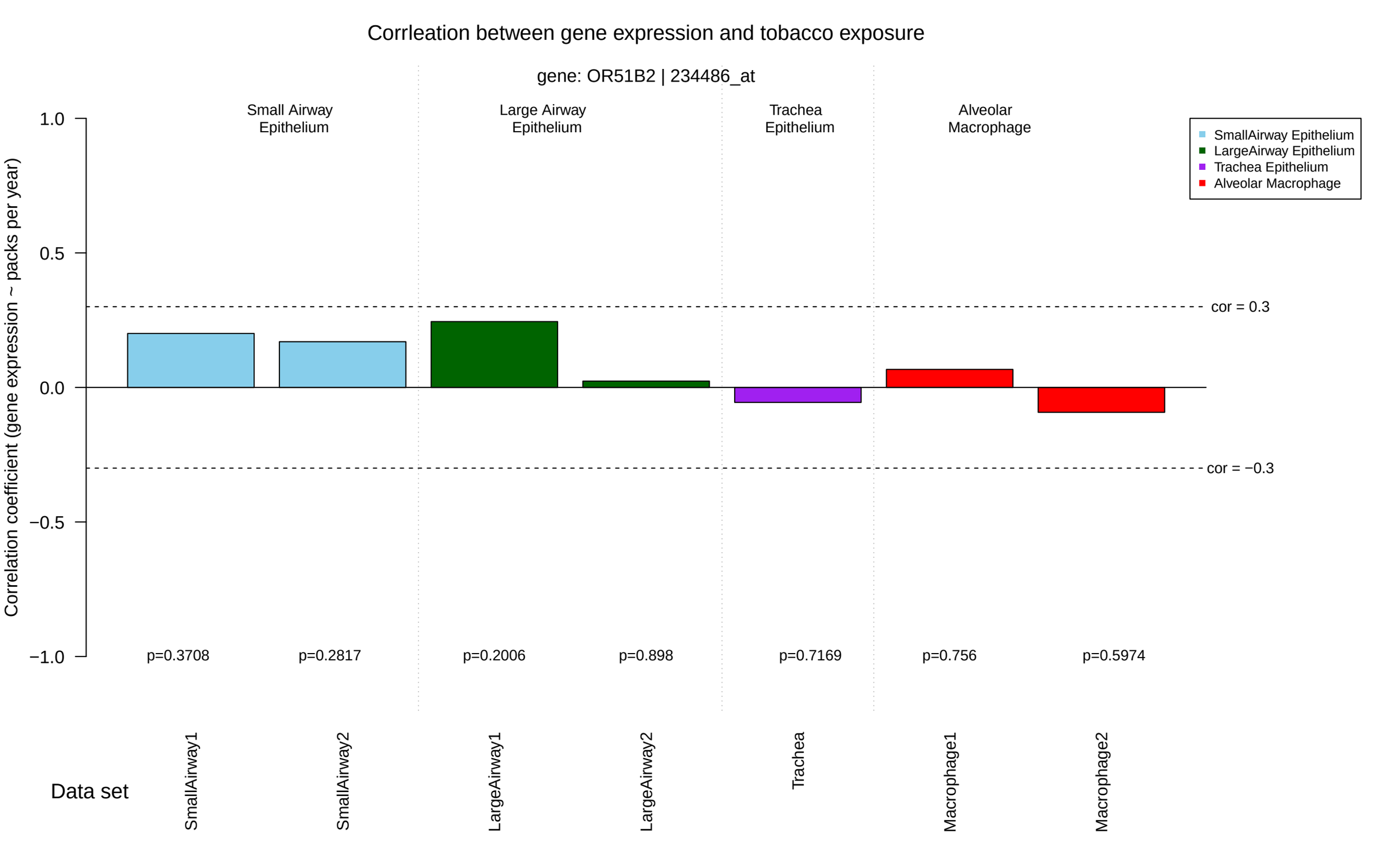

set: LargeAirway1 (p=0.0763)
OR51B2 | 234486_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4542)
OR51B2 | 234486_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9377)
OR51B2 | 234486_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3976)
OR51B2 | 234486_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6931)
OR51B2 | 234486_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OR51B2 | 234486_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3708
p=0.2817
p=0.2006
p=0.898
p=0.7169
p=0.756
p=0.5974
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2